

Ch. 8 — P02 (K) SimPy (s,S) Inventory Model

Problem Bank | Chapter 8 | Type: K | Difficulty:

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Objective

Implement a SimPy (s, S) inventory model and analyze cost-optimal policy parameters.

System Description

- Demand arrives as a Poisson process with rate $\lambda_d = 3$ units/day.
- Each demand has size Exponential(mean= 2 units).
- Replenishment lead time: Exponential(mean= 2 days).
- Holding cost: $h = \$1$ per unit per day.
- Backorder cost: $b = \$8$ per unit per day.
- Capacity: $S \leq 60$ units.

Tasks

1. **Implement** the `SSInventory` class (or use `simdes.models.inventory.SSInventory`). Run a single 1,000-day replication with $(s, S) = (10, 40)$ and report:
 - Average inventory level
 - Average backorder level
 - Average total cost per day
2. **Sweep** the reorder point $s \in \{5, 8, 10, 12, 15\}$ with $S = 40$ fixed. Run 20 replications per configuration. Plot mean total cost vs. s with 95% CIs.
3. **2D sweep:** For $s \in \{5, 10, 15\}$ and $S \in \{25, 35, 45\}$, run 15 reps each. Present results in a table and identify the (s, S) pair with lowest mean cost.
4. **Discuss:** Why does increasing s (while holding S fixed) reduce backorder cost but increase holding cost? Is there a unique optimal s^* ?

Submission

Jupyter notebook with model code, sweep plots, results table, and ≤ 300 -word discussion.